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3.1 Introduction to Unsupervised Learning

3.1.1 Definition

Unsupervised Learning is a type of machine learning in which:

- The model is trained on **unlabeled data** (no predefined output).
- The system tries to **identify patterns, structures, or relationships** within the data on its own.

“The algorithm learns from data without knowing the correct answers.”

Example

- Given customer data (age, income, purchase history), the system groups similar customers together without being told the categories.

3.1.2 Purpose of Unsupervised Learning

1. Pattern Discovery

Finding hidden relationships in data

Example:

In a supermarket dataset, the system discovers:

- People who buy **milk** also buy **bread**

2. Data Grouping (Clustering)

Group similar data points

Example:

An e-commerce website groups customers:

- Frequent buyers
- Occasional buyers
- New customers

3. Dimensionality Reduction

Reduce number of features

Example:

A dataset has 50 features → reduced to 5 important features using PCA

- Helps in faster computation and visualization

4. Anomaly Detection

Detect unusual data

Example:

- A sudden transaction of ₹1,00,000 from a normally low-spending account → flagged as fraud

5. Data Compression

Reduce storage size

Example:

- Image size reduced without losing much quality

3.1.3 Unsupervised Learning Approaches

Unsupervised learning is mainly divided into the following approaches:

1. Clustering

Clustering groups similar data points based on certain features.

Algorithms

- **K-Means Clustering**
- Hierarchical Clustering
- DBSCAN

How it Works

- Data points are grouped based on similarity (distance measures like Euclidean distance).

Example

- Customer segmentation in marketing

2. Association Rule Learning

Used to find relationships between variables in large datasets.

Algorithms

- Apriori Algorithm
- FP-Growth Algorithm

Example - Market Basket Analysis: "Customers who buy bread also buy butter"

3. Dimensionality Reduction

Reduces the number of input variables while retaining important information.

*Popular Techniques***i) PCA (Principal Component Analysis)**

PCA is a technique used to **“Reduce a large number of features into a smaller number while keeping most of the important information.”**

ii) t-SNE (t-Distributed Stochastic Neighbor Embedding)

t-SNE is a technique used in machine learning to **visualize high-dimensional data in 2D or 3D.**

t-SNE helps us:

“Take complex data with many features and convert it into a simple graph that we can see and understand.”

Purpose

- Simplify data
- Improve visualization
- Reduce computation time

4. Anomaly Detection

Identifies rare or unusual data points.

Example

- Fraud detection in banking
- Network intrusion detection

3.1.4 Applications and Scenarios of Unsupervised Learning

1. Customer Segmentation

- Group customers based on behavior
- Used in marketing strategies

2. Market Basket Analysis

- Identify purchase patterns
- Used in retail (e.g., Amazon recommendations)

3. Recommendation Systems

- Suggest products, movies, or music
- Based on user behavior patterns

4. Image Compression

- Reduce image size without losing quality

5. Anomaly Detection

- Detect fraud transactions
- Identify network attacks

6. Social Network Analysis

- Detect communities or groups within networks

7. Document Clustering

- Group similar documents/articles
- Used in search engines

8. Bioinformatics

- Gene clustering
- Disease pattern identification

Key Differences: Supervised vs Unsupervised Learning

Feature	Supervised Learning	Unsupervised Learning
Data	Labeled	Unlabeled
Goal	Predict output	Find patterns
Example	Classification, Regression	Clustering
Output	Known	Unknown

Advantages of Unsupervised Learning

1. No Need for Labeled Data

- Unsupervised learning works with **unlabeled data**
- No need for manual labeling (which is time-consuming and costly)

Example:

A company has customer purchase data but no categories → the algorithm automatically groups customers.

2. Finds Hidden Patterns

- Can discover **unknown patterns and relationships** in data
- Useful for **exploratory data analysis**

Example:

Supermarket data reveals:

- Customers who buy **milk also buy bread**

3. Useful for Data Exploration

- Helps understand data before applying other algorithms
- Gives insights into structure and distribution

Example: Before building a prediction model, data is grouped into clusters to understand behavior.

4. Works with Large Data

- Can handle large amounts of raw data
- Useful when labeling is impractical

Example: Social media platforms analyze millions of user interactions to identify trends.

5. Helps in Dimensionality Reduction

- Reduces number of features while keeping important information

Example: Using PCA, 50 features reduced to 5 → faster processing

6. Identifies Anomalies (Outliers)

- Detects unusual or rare data points

Example:

- Fraud detection in banking
- Detecting abnormal network activity

7. Cost-Effective

- No need to hire experts for labeling data
- Saves time and resources

Example: Medical datasets can be analyzed without labeling every record manually.

8. Improves Decision Making

- Helps businesses make better strategies based on discovered patterns

Example: E-commerce companies:

- Group customers → Targeted marketing

Question – What are limitations of Limitations of Unsupervised Learning [5 Marks]

Limitations of Unsupervised Learning

1. No Correct Output (No Labels)

The model does not have predefined answers, so it cannot know if the result is correct or not.

2. Difficult to Interpret

The algorithm forms groups or patterns, but understanding what those groups actually represent is not always easy.

3. No Direct Accuracy Measure

Since there are no labels, we cannot calculate accuracy like in supervised learning.

4. Sensitive to Data Quality

If the data has noise, missing values, or errors, the results may be incorrect or misleading.

5. Results Depend on Algorithm

Different algorithms (like K-Means or DBSCAN) may give different outputs for the same data.

6. Requires Human Expertise

A human expert is needed to interpret and give meaning to the results.

7. Computationally Expensive

Some techniques take more time and resources, especially for large datasets.

8. Risk of Incorrect Grouping

Choosing the wrong number of clusters can lead to poor or confusing results.